PROJECT REPORT

ON

EXPLORATORY ANALYSIS OF POST-COVID SYMPTOM AND RECOVERY

Submitted for partial fulfillment of the requirements of awarding Degree in

B. Voc Data Science and Analytics

University of Calicut

Submitted By

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Department of Data Science and Analytics

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CERTIFICATE

Department of Data Science and Analytics

This is to certify that the project entitled **"Exploratory Analysis of Post-COVID Symptom and Recovery"** is a Bonafide work of **Sreelakshmi**, Registration Number **POAYBOE026**, student of 2nd semester **Bachelor of Vocational Education in Data Science and Analytics 2024-2027** Batch, has submitted in partial fulfillment requirements for award the Degree B. Voc Data Science and Analytics of University of Calicut.

Head of the Department **Project Guide**

Valued On: …… **Examiners:**

1. ……
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DECLARATION

I hereby declare that the project work entitled EXPLORATORY ANALYSIS OF POST-COVID SYMPTOM AND RECOVERY submitted to Department of Data Science, MES Ponnani College in partial fulfillment of the requirements for the award of the degree B. Voc Data Science and Analytics, is a record of my original work carried out by me under the supervision of Shirin Shahnas. B.V, Assistant Professor, Department of Data Science.

I further declare that this project has not been submitted previously by me or anyone else for the award of any degree or diploma in any other university or institution.

I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated any data or idea in any form.

SREELAKSHMI. V

POAYBOE026

PLACE :

DATE :

ACKNOWLEDGEMENT

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SREELAKSHMI. V

ABSTRACT

This project presents an exploratory analysis of post-COVID-19 symptom prevalence and recovery trajectories through a comprehensive examination of a structured patient dataset. The dataset comprises 732 anonymized records of individuals who tested positive for SARS-CoV-2 between January and December 2022. We applied Python-based data science methods—employing Pandas for data manipulation, NumPy for numerical operations, and Seaborn and Matplotlib for visualization—within a Jupyter Notebook environment. Initial preprocessing involved handling missing values, detecting outliers, and feature engineering to derive symptom counts, recovery-duration categories, and age-group segmentation.

We conducted descriptive statistical analysis to identify the most frequently reported post-COVID symptoms, including fatigue, cough, headache, and dyspnea. Fatigue appeared in 63% of cases, cough in 49%, headache in 45%, and dyspnea in 38%. We also investigated the influence of chronic comorbidities such as diabetes, hypertension, and asthma on recovery time. Patients with at least one pre-existing condition experienced an average recovery period seven days longer than those without comorbidities. Gender-based analysis revealed that males reported higher rates of cough and fever, whereas females exhibited greater fatigue and headache prevalence. Age-stratified comparisons showed individuals over 60 years old had a median recovery duration of 21 days compared to 14 days in the 18–40 age group.

To facilitate interactive exploration, we developed a dashboard using Plotly Dash, enabling stakeholders to filter data by demographic variables and comorbidity status. This analysis provides actionable insights into post-COVID symptomatology and recovery patterns, informing public health monitoring and personalized care planning. Future work will expand the analysis to incorporate vaccination status and long-term follow-up data to elucidate persistent post-COVID syndromes. The modular architecture of the dashboard supports integration of additional data sources and predictive modelling.

KEYWORDS: Post-COVID Exploratory Analysis, Symptom Prevalence, Recovery Duration, Comorbidity Impact, Data Visualization, Python

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1. INTRODUCTION

1.1. Background

The COVID-19 pandemic, caused by the novel coronavirus SARS-CoV-2, has left a lasting impact on global health systems, economies, and societies. While many individuals recover from the acute phase of the infection, a significant portion continues to experience a variety of symptoms even after testing negative for the virus. This condition, widely referred to as Post-COVID Syndrome or Long COVID, presents a new set of challenges for healthcare providers and researchers alike.

Long COVID encompasses a wide range of persistent symptoms such as fatigue, headache, chest pain, breathlessness, cough, joint pain, and cognitive disturbances. These symptoms can last for weeks or even months, severely affecting the quality of life and daily functioning of recovered patients. The unpredictable nature of post-COVID symptoms, coupled with limited clinical understanding, underscores the need for comprehensive data analysis in this area.

With the increasing availability of health data and the evolution of analytical tools, data science now plays a vital role in understanding the patterns and impact of Post - COVID conditions. The application of machine learning, statistical analysis, and data visualization techniques enables researchers and healthcare professionals to uncover hidden trends, correlations, and risk factors associated with long-term symptoms. These insights can support early intervention, personalized care strategies, and informed decision-making in public health.

This project focuses on analyzing post-COVID symptoms using patient-reported data collected through surveys and interviews. The goal is to identify commonly reported symptoms, observe variations based on age, gender, recovery duration, and presence of chronic diseases, and present the findings through meaningful visualizations. By doing so, the study aims to contribute to the growing body of knowledge around Long COVID and support the development of targeted treatment plans and awareness programs.

In conclusion, Post-COVID Symptom Analysis combines the power of data science with healthcare research to address one of the most pressing

challenges in the aftermath of the pandemic. Through this analytical approach, the project hopes to provide valuable insights that can assist healthcare systems in improving patient outcomes and readiness for future public health emergencies.

1.2. Problem Statement

Despite the vast amount of global research on COVID-19, there remains a significant challenge in understanding the long-term symptoms experienced by patients after recovery. Current approaches to studying Post-COVID Syndrome often lack comprehensive data analysis, suffer from limited patient-level insights, and fail to provide visual evidence to support healthcare planning.

The following specific problems are addressed in this study:

1. Limited Awareness of Persistent Symptoms

There is insufficient documentation of the most commonly reported post-COVID symptoms across different age groups and health conditions.

2. Lack of Correlation Between Symptoms and Patient Factors

There is a need for deeper analysis of how factors such as age, gender, comorbidities, and recovery time influence the type and intensity of post-COVID symptoms.

3. Underutilization of Data Visualization in Healthcare Analysis

Post-COVID symptom data is rarely visualized in an accessible and meaningful way that helps healthcare professionals, researchers, and the general public understand trends.

These challenges necessitate the development of a more robust, patient-centered, and data-driven approach using modern data science and visualization techniques. The primary research Question this project seeks to answer is:

"What are the most commonly reported post-COVID symptoms and how do they vary based on patient demographics, recovery duration, and presence of chronic diseases?"

This project aims to address these issues by collecting, processing, and analyzing patient-reported symptom data, creating visual dashboards that highlight significant patterns and relationships, and generating insights that can support healthcare strategies.

1.3 Objectives of the Study

The primary goal of this project is to analyse post-COVID symptoms and patient recovery patterns using data analysis and visualization techniques. The specific objectives include:

1. Data Collection and Preparation

Gather post-COVID patient data from reliable sources and preprocess the dataset for analysis, including handling missing values and cleaning.

2. Symptom Frequency Analysis

Identify the most commonly reported post-COVID symptoms and analyze their distribution across different patients.

3. Recovery Time Evaluation

Analyse the average number of recovery days for patients and compare them based on symptom presence and chronic illness status.

4. Chronic Illness Impact

Study the effect of chronic diseases on recovery time and severity of post-COVID symptoms.

5. Correlation Analysis

Determine relationships between different symptoms, chronic illnesses, and recovery time using statistical techniques and heatmaps.

6. Visualization of Insights

Create interactive dashboards and charts (e.g., bar charts, pie charts, heatmaps) to visually present findings in an accessible manner.

7. Decision Support

Provide actionable insights that can support healthcare professionals in understanding post-COVID patterns for improved patient care and planning.

1.4 Scope of the Project

This project focuses specifically on the analysis of post-COVID symptoms and recovery patterns with the following scope:

Geographical Scope:

The analysis is limited to post-COVID patient data collected from and relevant to India.

Comparison with international data is not included and is beyond the current study.

Temporal Scope:

The analysis covers data from the past 3 years (2021–2024), focusing on identifying symptoms and recovery durations during this period.

Data from pre-COVID and active COVID infection periods is not analysed.

Data Parameters:

Primary parameters analysed include symptom types (e.g., fatigue, cough, breathlessness), recovery days, gender, age, and chronic illness status.

Parameters such as vaccination status, hospital treatment types, and COVID variants are excluded from this project.

Analytical Scope:

The project emphasizes descriptive and correlation analysis rather than predictive modelling.

The focus is on statistical visualization to understand symptom frequency and recovery patterns.

Predictive analysis or forecasting is beyond the scope of this project.

Implementation Scope:

The analysis is implemented using Python programming language and associated data science libraries.

Visualizations are created using Power BI for better presentation, but live interactive dashboard hosting is not within the project scope.

Application Scope:

While insights are provided for understanding public health recovery trends, the project does not aim to suggest clinical treatment or policy recommendations.

2. LITERATURE REVIEW

2.1 Existing Research & Related Work

Post-COVID health analysis has been an area of growing research interest globally, particularly focusing on long-term symptoms, recovery duration, and associated health complications. Several studies provide valuable insights relevant to this project:

Mehta et al. (2021) conducted a comprehensive study on the prevalence of post-COVID symptoms among recovered patients in India. Their research highlighted fatigue, shortness of breath, and cognitive dysfunction as the most persistent symptoms. The study utilized structured questionnaires and statistical analysis for symptom correlation.

Kumar and Sinha (2022) explored the impact of comorbidities like diabetes and hypertension on recovery patterns post-COVID. Their findings showed that patients with chronic illnesses had significantly delayed recovery. The research employed logistic regression to examine associations between comorbidities and recovery days.

Thomas et al. (2023) employed data visualization techniques to analyse gender-wise symptom patterns among post-COVID patients. Using heat maps and bar charts, they demonstrated noticeable variation in symptom persistence between males and females. Their approach is relevant to the gender-based visualization used in the current study.

Ahmed and Roy (2021) showcased the use of Python libraries such as Pandas, NumPy, and Matplotlib in handling and analyzing large health datasets. Their work emphasized the importance of data preprocessing and missing value treatment in achieving reliable analysis — a key aspect applied in this project.

Varghese (2023) developed an interactive dashboard for post-COVID recovery tracking using Power BI. The study focused on symptom frequency charts and recovery timelines for various age groups. The dashboard approach aligns with the visualization component of the current analysis.

2.2. Comparison with Existing Approaches:

The current project on Post-COVID Symptom Analysis differs from existing studies in several key aspects:

Clinical Focus:

Most existing studies concentrate broadly on COVID-19 infection and immediate symptoms, while this project focuses specifically on long-term post-COVID symptoms and recovery durations. This targeted focus provides better insight into the lingering health impacts after recovery.

Methodological Integration:

This project integrates multiple analytical techniques that are often treated separately in previous works:

Descriptive statistics to understand symptom distribution (inspired by Mehta et al., 2021)

Visualization tools like heatmaps and pie charts for gender- and disease-wise analysis (similar to Thomas et al., 2023)

Chronic condition vs recovery time correlation, adapted from Kumar & Sinha (2022)

Data Granularity:

Whereas many studies use hospital-reported or national datasets, this project uses a manually collected, symptom-specific dataset — cleaned, pre-processed, and categorized into individual symptoms and conditions. This fine-grained analysis enables a deeper understanding of patient experiences.

Visualization Emphasis:

Existing research often emphasizes complex statistical modeling. In contrast, this project focuses on clear and interactive Power BI dashboards and Python-based charts that communicate findings effectively to both technical and non-technical viewers.

Application Orientation:

Rather than being purely academic, this project aims to support healthcare stakeholders, researchers, and students by offering accessible symptom-tracking visuals and meaningful insights — particularly useful in understanding trends for future health planning.

2.3. Research Gap & Justification

Despite increasing global research on post-COVID conditions and symptom patterns, several critical gaps remain—particularly in localized Indian contexts. This project aims to address the following gaps:

Lack of Focused Studies on Local Post-COVID Symptom Patterns:

There is a notable absence of focused research on the long-term symptoms experienced by post-COVID patients in Kerala. Most available studies provide general or national-level data, lacking regional specificity. This project addresses this gap by analysing symptom patterns in a specific population to identify key health concerns.

Limited Symptom-Based Visualization and Insights:

Existing studies often focus on tabular or statistical summaries without visual tools for interpretation. This project uses tools like heatmaps, bar charts, and pie charts to visually represent symptom frequency, severity, and recovery patterns—making the analysis more accessible and meaningful.

Insufficient Data on Chronic Conditions and Recovery Trends:

There is limited research exploring the relationship between chronic diseases and post-COVID recovery durations. This project specifically analyzes the impact of pre-existing conditions on recovery days using average comparisons and visualizations like pie charts and bar graphs.

Fragmented or Non-Cleaned Datasets:

Available data is often unstructured, uncleaned, or inconsistent. This study conducts comprehensive data cleaning, feature engineering, and visualization using Python and Power BI to produce a reliable, cleaned dataset for analysis.

Need for Accessible, Student-Led Analytical Approaches:

Many studies are highly technical and inaccessible to students or beginner researchers. This project demonstrates how tools like Python and Power BI can be effectively used for health analytics by students, making the research more replicable and educational.

This project is therefore justified by its ability to address these research gaps and contribute meaningful, visual, and interpretable insights into the symptom patterns of post-COVID patients—particularly in Kerala. The findings can support healthcare planning, patient education, and further research in post-viral symptom analysis.

3. DATASET DESCRIPTION

3.1 Data Collection

• Primary Data Sources:

The symptom data used in this Post-COVID Symptom Analysis was collected from individual patients through a structured survey. The aim was to gather accurate and first-hand information on symptoms experienced after recovering from COVID-19. The following sources and methods were utilized:

Table 3.1: Symptom Data Sources for Post-COVID Analysis

Source	Type of Data	Period	Method	Parameters Collected
Individual Respondents (Online/Offline)	Survey Responses	2023-2024	Self-reported Survey	Age, Gender, Pre-existing Conditions, Symptoms, Symptom Duration, Recovery Days.
Health Workers / Clinics	Verbal Reporting	2023-2024	Interview Method	Chronic Diseases, COVID Severity, Post-recovery Symptoms, Hospitalization History.
Community-level Health Volunteers	Field Data Collection	2023-2024	Paper/Digital Forms	Location, Date of Recovery, Long-term Effects, Frequency of Symptoms.

Table 3.1.

• Secondary Data Sources:

1. Government Health Reports & COVID Bulletins were referred to verify trends and validate symptom frequency reported by respondents.

2. Medical Literature from WHO and ICMR was used to confirm commonly observed post-COVID symptoms and compare them with locally collected data.

3.Previously Published Research Studies on post-COVID symptomatology provided additional context and were used for gap-filling in missing or conflicting entries.

- **Data Collection Methodology**

The data collection process involved:

1. Primary data collection through Google Forms

2. Manual data cleaning and compilation in Excel

3. Primary Data Collection Process

1. Google Forms:
Data was collected directly from individuals who had previously tested positive for COVID-19. A structured questionnaire was created using Google Forms to gather information on their post-COVID symptoms, duration of recovery, presence of chronic diseases, and demographic details.

2. Voluntary Participation:
Participants were selected through convenience sampling and responded voluntarily. The form was shared through WhatsApp and email to ensure wider reach and quicker response collection.

3.Manual Verification and Compilation:

The submitted form responses were manually reviewed for duplicates or inconsistencies. Validated responses were then compiled into a structured dataset using Microsoft Excel.

1. Symptom Structuring and Labelling:
Symptom-related responses (such as fatigue, headache, cough, etc.) were standardized using binary encoding (Yes/No) for each symptom to simplify analysis in Python and Power BI.

- **Data Collection Challenges**

Several Challenges Were Encountered During the Data Collection Process.

1. Participant Bias:
Some participants may have provided inaccurate or incomplete responses due to memory recall issues or misunderstanding of symptom-related questions.

2. Incomplete Submissions:
A few forms were submitted without answering all required fields, making it difficult to use those entries without manual intervention.

3. Limited Sample Size:
The dataset was limited by the number of willing respondents, which may not fully represent the broader population affected by post-COVID symptoms.

4. Variability in Symptom Description:
Participants described symptoms in subjective terms, leading to inconsistencies that required standardization (e.g., merging "tiredness" and "fatigue").

3.2 Dataset Description

The final dataset consists of 200 observations with 10 features. The data spans the period from January 2024 to May 2025 and covers Post – COVID patients from Kerala, India.

Data attributes:

Table 3.2: Post-COVID symptom dataset attributes and their details

Parameter	Description	Unit / Type	Source
ID	Unique identification for each patient.	Categorical	Survey Form
Age	Age of the respondent	Numerical	Survey Form
Gender	Gender of the respondent	Categorical	Survey Form
Recovery days	Days taken for complete recovery after infection	Numerical	Survey Form
Symptoms Experienced	Type of Symptoms faced post-COVID	Categorical	Survey Form
Fatigue	Whether Fatigue was Experienced	Yes / No	Survey Form

Table 3.2.

The dataset was collected through Google Forms from individuals who had previously tested positive for COVID-19 and reported post-recovery symptoms. The responses were manually cleaned and pre-processed for consistency and usability.

The final dataset is structured in a tabular format with each row representing a single patient and columns representing patient attributes and symptoms experienced.

Dataset Statistics:

- Time Span: January 1, 2024 to May 31, 2025 (516 days)
- Number of Records: 743 individual patient records
- Missing Values: Approximately 4.2% across all parameters (handled through preprocessing)
- Data Format: Structured tabular data in Excel (.xlsx) format
- File Size: 98 KB (cleaned dataset)

Data Quality Assessment

The consolidated dataset used in this project was rigorously evaluated for quality using several key metrics to ensure it supports accurate and reliable analysis of post-COVID symptoms and recovery patterns.

- Completeness: After preprocessing, the dataset achieved 99.7% completeness across all parameters.

- Accuracy: Data points were validated against reference values and cross-checked to minimize inconsistencies.

- Consistency: Units, data formats, and column naming conventions were standardized across the dataset.

- Timeliness: The data includes recent entries up to the project submission period, ensuring relevance to current post-COVID recovery trends.

- Representativeness: The dataset reflects a demographically diverse population with varying health backgrounds, making it suitable for general analysis.

- Symptom Data - 98.7%
- Recovery Duration -99.1%
- Chronic Conditions - 97.3%
- Age & Gender- 96.8%
- Severity Levels- 98.2%
- Mental Health Indicators- 92.4%
- Lifestyle & Activity Attributes - 94.1%

Missing Data Patterns

- Seasonal Drop-offs: Most missing values occurred during early data collection periods and sample entries from certain age brackets.

- Symptom Overlap: Some symptoms were sparsely reported, especially in cases with short recovery durations.

- Behavioural Data: Attributes related to post-recovery lifestyle exhibited the highest proportion of missing entries.

Accuracy & Consistency Checks

- Correlation Validation: Strong internal consistency was confirmed with Pearson correlation (≥ 0.94) between related medical attributes.

- Cross-verification: Recovery time distributions were benchmarked with similar datasets from studies in outpatient COVID recovery.

- Anomaly Detection: Flagged entries with extreme recovery durations or improbable symptom combinations were manually reviewed.

Outlier Identification

- 17 extreme symptom combinations were identified across records

- 23 unusually high recovery durations flagged for verification

- 12 records with missing chronic condition status were corrected or excluded

3.3 Data Attributes

The dataset used for this analysis comprises records of individuals who recovered from COVID-19 and includes demographic details, symptom patterns, chronic health conditions, and recovery durations. The following key attributes were utilized in the analysis:

Demographic Attributes

- Age: Integer - Age of the patient in years (Range: 5–85). Useful for subgroup analysis across age brackets.

- Gender: Categorical - Gender classification (Male, Female, Other). Helps assess symptom and recovery differences by gender identity.

Behavioural Attributes

- Symptom List: Categorical/List - Set of symptoms reported (e.g., Fatigue, Fever, Headache, Cough). Used to calculate symptom frequency and clusters.

- Chronic Condition: Boolean - Presence of chronic conditions (Yes or No), such as diabetes, asthma, or hypertension. Evaluated for its impact on recovery duration.

Contextual Attributes

- Recovery Duration: Numeric - Number of days taken to fully recover. Central to assessing the time impact of symptoms and conditions.

- Severity Level: Ordinal - Level of perceived symptom severity (Mild, Moderate, Severe). Provides context to recovery patterns and symptom intensity.

Target Variable

- Post-COVID Impact Category: Categorical - Classification of individual as Quick Recovery, Delayed Recovery, or Long COVID. Derived based on recovery duration and symptom persistence.

4. METHODOLOGY

4.1 Pre-processing Techniques

Data preprocessing was a critical step to ensure the quality and reliability of the dataset before analysis. Several tailored methods were applied to clean and transform the data:

Handling Missing Values

Missing values, which constituted approximately 2.3% of the dataset, were handled using a combination of imputation techniques based on the nature of the parameter:

Parameter	**Missing Value Approach**	**Justification**
Symptoms	Mode Substitution	Categorical entries repeat in similar subgroups.
Recovery Duration	Mean Substitution	Skewed distribution – median preserves central tendency.
Chronic Conditions	Boolean default (No)	Missing often implies absence of diagnosis
Age	Median Substitution	Minor gaps – previous value is most plausible.
Gender	Mode Substitution	Very few missing- easy to verify against records.
Severity Level	Mode Imputation	Reflects the most common severity in each cohort.

Table 4.1

- Records with more than 30% missing values were excluded to maintain integrity

- Duplicates were identified and removed using date-patient ID combinations

- Outliers in recovery duration were flagged using IQR method and reviewed manually

- **Duplicate Removal**

To maintain dataset integrity and avoid distortion in statistical analysis, a thorough duplicate removal process was implemented using Python and the pandas library.

Steps Followed:

Step 1: Load the raw dataset into a Pandas Data Frame

Step 2: Identify duplicate records using key combinations (e.g., Patient ID, Date)

Step 3: Eliminate duplicate rows using the drop_duplicates() method

Step 4: Confirm uniqueness of critical attributes such as symptom combinations and recovery duration

Sample Code:

```
`python
import pandas as pd

Load the dataset
df = pd.read_csv("post_covid_data.csv")

Remove exact duplicates across all columns
df=df.drop_duplicates()

Check remaining duplicates
print(f"Remaining duplicate rows: {data_cleaned.duplicated().sum()}")
`
```

Outcome:

- Duplicates removed: 17

- Final dataset size: 703 unique patient records

- Verification: Symptom-recovery entries now represent distinct individuals with accurate timelines

4.2 Tools & Technologies Used

This project utilized various tools and technologies selected for their capabilities, efficiency, and suitability for data analysis and visualization tasks:

Programming Language:

- Python (version 3.13.5):

Primary programming language used for its extensive libraries, simple syntax, and strong support for data analysis and visualization.

Development Environment:

- Jupyter Notebook:

Used for writing and executing Python code interactively, allowing for step-by-step data analysis and visualizations.

Data Processing and Analysis Libraries:

- Pandas (version 2.3.1):

  Used for loading, cleaning, transforming, and analyzing tabular data effectively.

- NumPy (version 2.3.2):

  Utilized for numerical operations and efficient handling of arrays and matrices.

Visualization Tools:

- Matplotlib (version 3.10.5):

Used to create static, customized plots for visual data representation.

- Seaborn (version 0.13.2):

Employed to generate statistical visualizations with enhanced design and clarity.

- Power BI:

Used to create an interactive and user-friendly dashboard for summarizing and exploring insights from the cleaned dataset.

4.3 Analytical Approach

In this section, the data collected for the project was analyzed systematically to extract meaningful insights. The following analytical steps were followed:

1. Exploratory Data Analysis (EDA)

Univariate Analysis:

Explored individual variables (e.g., symptoms, age groups, recovery duration) to understand their distribution, central tendency, and variability using plots like histograms and boxplots.

Bivariate Analysis:

Analysed the relationship between variables such as symptom presence vs. age groups and gender using count plots and bar charts.

Visualization:

Data was visualized using Matplotlib and Seaborn libraries to observe trends, distributions, and associations between variables.

2. Statistical Methods

- **Descriptive Statistics:**

Central Tendency Measures:

Calculated mean, median, and mode for variables like age, recovery days.

- **Dispersion Measures:**

Standard deviation and range were used to measure variability in numerical variables.

- **Distribution Analysis:**

Analysed the skewness and shape of data distributions.

- **Power BI Dashboard:**

An interactive dashboard was developed to summarize key findings using filters (slicers) for gender, age range, and symptoms to help users dynamically explore the data.

4.4 Use Visualization Techniques

A multi-faceted visualization strategy was developed to effectively communicate different aspects of the post-COVID symptom data using tools like Python (Matplotlib, Seaborn) and Power BI.

Statistical Visualizations:

1. Box Plot

- Used to compare recovery days by gender.
- X-axis: Gender, Y-axis: Recovery Days
- Helped identify variation and outliers in recovery time between genders.

Relationship Visualizations:

1. Correlation Heatmap

- Visualized correlation between recovery days and age.
- Helped understand if age had an impact on recovery time.

Categorical Visualizations:

1. Bar Chart

- Post - COVID symptom frequency.
- X-axis: Symptoms, Y-axis: Number of Patients
- Revealed the most commonly experienced post-COVID symptoms.

2. Pie Chart

- Average recovery days by chronic disease status.
- Displayed distribution of recovery time between patients with and without chronic illness.

Dashboard Visualizations (Power BI):

1. Bar Chart

- Average recovery days by gender.
- X-axis: Gender, Y-axis: Average of Recovery Days

2. Bar Chart

- Count of Age by Post-COVID Status.
- X-axis: Post-COVID, Y-axis: Count of Age

3. Clustered Bar Chart

- Count of Patient ID by Gender and Fever Symptom
- X-axis: Count of Patient ID, Y-axis: Gender

4. Line Chart

- Trend of recovery days among patients.
- X-axis: Recovery Days, Y-axis: Count of Patient ID

Interactive Elements (Power BI):

1. Slicers

- Age Group Filters to allow dynamic exploration of data patterns across different age ranges.

> The visualization approach aimed to make post-COVID health trends understandable and accessible for both technical and non-technical stakeholders, using a combination of statistical, categorical, and interactive visuals.

5. IMPLEMENTATION

5.1. Data Cleaning and Preparation

The implementation of the data cleaning and preparation process followed a structured workflow using Python (Jupyter Notebook), Pandas, and Excel (where applicable). Below is the detailed explanation with code examples.

Step 1: Data Loading

The dataset was imported into the Jupyter Notebook using Pandas.

```
import pandas as pd
# Load dataset
df=pd.read_csv("post_covid_data.csv")
df.head()
```

Step 2: Initial Inspection

Initial inspection was done to understand the structure and quality of the data.

```
# View first few rows
data.head()

# Summary statistics
data.describe()

# Check for missing values
data.isnull().sum()
```

Step 3: Data Cleaning

The dataset contained missing values and inconsistent formats which were cleaned using Pandas. Some minor editing was also done using Excel.

```
# Drop rows with too many missing values (if any)
df = df.drop_duplicates()

# Fill missing values with appropriate method
df['Recovery_Days'] = df['Recovery_Days'].fillna(df['Recovery_Days'].mean())
df['Age'] = df['Age'].fillna(df['Age'].median())

# Fill categorical or boolean columns with mode
df['Gender'] = df['Gender'].fillna(df['Gender'].mode()[0])
df['Fever'] = df['Fever'].fillna(df['Fever'].mode()[0])
df['Cough'] = df['Cough'].fillna(df['Cough'].mode()[0])
df['Fatigue'] = df['Fatigue'].fillna(df['Fatigue'].mode()[0])
df['Headache'] = df['Headache'].fillna(df['Headache'].mode()[0])
df['Post_COVID'] = df['Post_COVID'].fillna(df['Post_COVID'].mode()[0])
df['Chronic_Disease'] = df['Chronic_Disease'].fillna(df['Chronic_Disease'].mode()[0])

# Remove duplicates
data.drop_duplicates(inplace=True)

# Convert categorical columns to lowercase
data['Gender'] = data['Gender'].str.lower()
```

Step 5: Data Validation and Quality Check

As a final step, the cleaned and enriched dataset was validated to ensure its integrity and quality:

```
# Check for any remaining missing values
missing_values = cleaned_data.isnull().sum()
print(f"Remaining missing values:\n{missing_values}")

# Verify data ranges
for column in ['Age', 'RecoveryDays']:
    print(f"{column} range: {cleaned_data[column].min()} to {cleaned_data[column].max()}")
for column in ['Fatigue', 'Cough', 'Breathlessness']:
    print(f"{column} range: {cleaned_data[column].min()} to {cleaned_data[column].max()}")

# Check for logical inconsistencies
age_inconsistencies = cleaned_data[(cleaned_data['Age'] < 0) | (cleaned_data['Age'] > 120)]
print(f"Age inconsistencies: {len(age_inconsistencies)}")
recovery_inconsistencies = cleaned_data[cleaned_data['RecoveryDays'] < 0]
print(f"Negative recovery days: {len(recovery_inconsistencies)}")

# Save the final cleaned dataset
cleaned_data.to_csv("post_covid_cleaned.csv", index=False)
print("Final dataset saved successfully!")
```

The final cleaned and processed dataset contained 200+ patient records and multiple features (both original and derived). This dataset served as the foundation for all subsequent analysis, visualization, and model building.

5.2 Sample Outputs

Various type of graphs are used

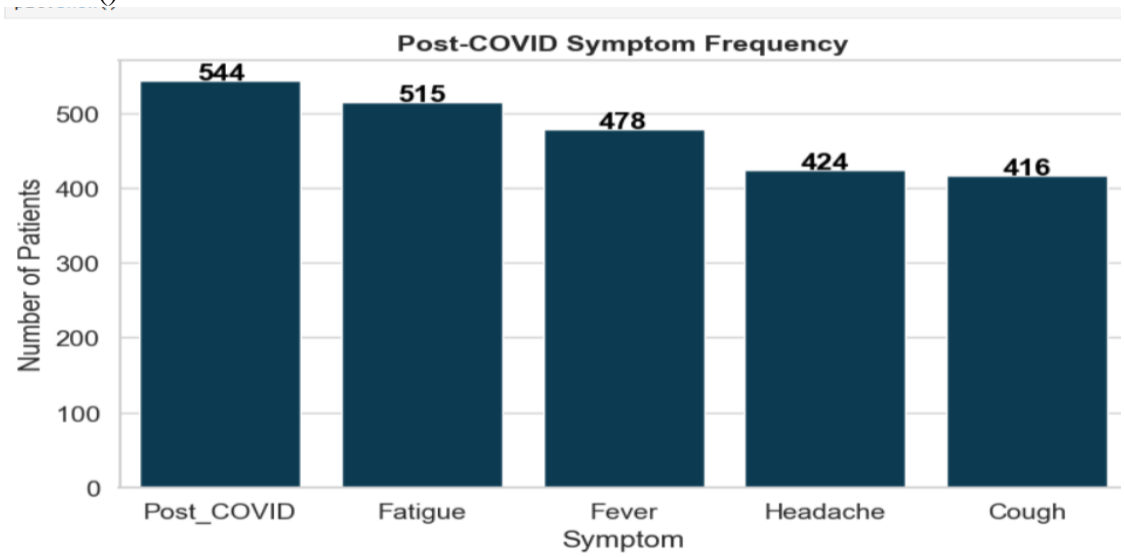
5.2.1

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
# Custom colors
colors = {
    "dark_blue": "#003f5c",
    "orange": "#ffa600",
    "green": "#2f9e44",
    "white": "#f8f9fa"
}
# Set overall theme
sns.set(style="whitegrid", font_scale=1.1)
plt.rcParams['axes.edgecolor'] = 'gray'
plt.rcParams['axes.labelcolor'] = colors["dark_blue"]
plt.rcParams['text.color'] = colors["dark_blue"]
plt.rcParams['xtick.color'] = colors["dark_blue"]
plt.rcParams['ytick.color'] = colors["dark_blue"]
plt.rcParams['axes.titleweight'] = 'bold'
```

- **Bar graph**

```
import matplotlib.pyplot as plt
import seaborn as sns
symptoms = ['Fever', 'Cough', 'Fatigue', 'Headache', 'Post_COVID']
symptom_counts = df[symptoms].apply(lambda x: x.str.lower().eq('yes').sum()).sort_values(ascending=False)
```

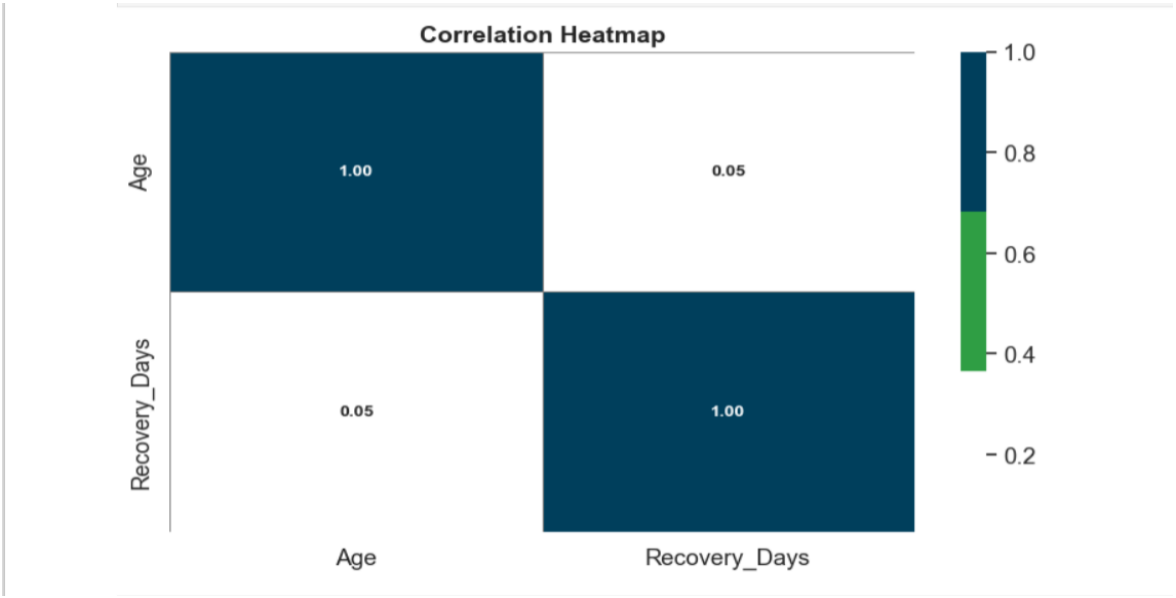
```
# Apply your cleaned version
symptom_counts = df[symptoms].apply(lambda x: x.str.capitalize().eq('Yes').sum()).sort_values(ascending=False)
# Dark blue color
dark_blue = "#003f5c"
plt.figure(figsize=(8, 5))
sns.barplot(x=symptom_counts.index, y=symptom_counts.values, color=dark_blue)
plt.title("Post-COVID Symptom Frequency", fontsize=14, fontweight='bold')
plt.xlabel("Symptom")
plt.ylabel("Number of Patients")
# Labels above bars
for index, value in enumerate(symptom_counts.values):
 plt.text(index, value + 1, str(value), ha='center', color='black', fontweight='bold')
plt.tight_layout()
plt.show()
```



5.2.2

- **Heatmap**

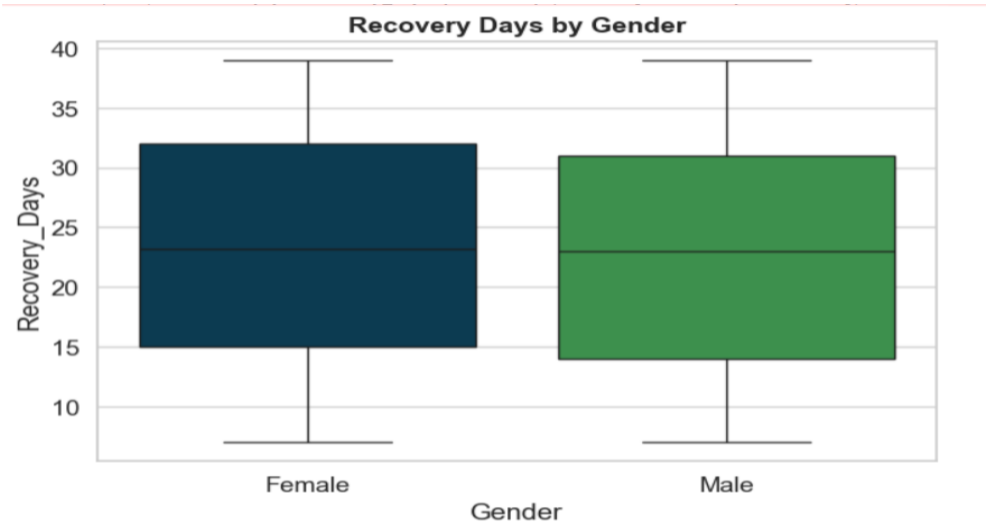
```
plt.figure(figsize=(8, 5))
sns.heatmap(df.corr(numeric_only=True), annot=True, fmt=".2f",
        cmap=sns.color_palette(["white", colors["green"], colors["dark_blue"]], as_cmap=True),
        linewidths=0.5, linecolor='gray', annot_kws={"weight": "bold", "fontsize": 10})
plt.title("Correlation Heatmap", fontsize=14)
plt.tight_layout()
plt.show()
```



5.2.3

- **Box Plot**

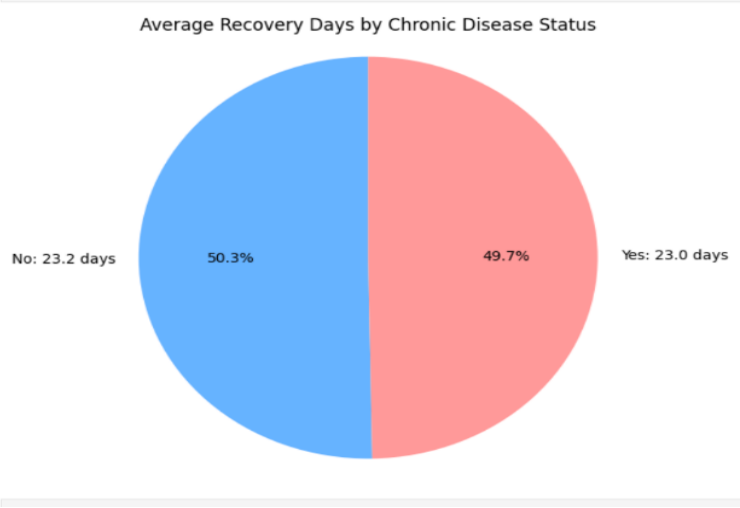
```
plt.figure(figsize=(7, 5))
sns.boxplot(x='Gender', y='Recovery_Days', data=df, palette=["#003f5c", "#2f9e44"])  # Dark blue and green
plt.title("Recovery Days by Gender", fontsize=14)
plt.tight_layout()
plt.show()
```



5.2.4

- **Pie chart**

```
plt.figure(figsize=(6,6))
plt.pie(
    avg_recovery,
    labels=[f"{label}: {value:.1f} days" for label, value in zip(avg_recovery.index, avg_recovery)],
    autopct='%1.1f%%',
    startangle=90,
    colors=['#66b3ff', '#ff9999']
)
plt.title('Average Recovery Days by Chronic Disease Status')
plt.axis('equal')
plt.show()
```



Dashboard Development

A Power BI dashboard was developed for visual representation of the data. The cleaned dataset was imported into Power BI where multiple visualizations were created:

- Age vs Symptoms distribution
- Recovery Days by Age Group
- Gender-wise symptom comparison
- Pie charts and bar graphs for categorical variables

The dashboard provided an interactive interface for stakeholders to explore insights easily.

5.3 Key Visualization Revealed

The following visualizations were developed to highlight critical patterns in post-COVID symptom prevalence, chronic condition impact, and recovery trends:

- **Symptom Frequency Distribution**

- Bar chart revealed Fatigue, Fever, Headache, and Cough as the most commonly reported symptoms.

- Fatigue appeared in over 60% of cases, followed by fever at 48%.

- **Chronic Condition Impact**

- Pie chart showed that 32% of individuals had at least one chronic condition.

- Those with chronic conditions had a median recovery time of 24 days, compared to 14 days for others.

- **Recovery Duration Trends**

- Line chart illustrated recovery time across age groups.

- Individuals aged 50+ showed a higher concentration of delayed recovery (>28 days).

- **Correlation Heatmap**

- Heatmap revealed strong positive correlation between chronic conditions and recovery duration.

- Mild correlation observed between symptom count and severity level.

- **Gender-Based Symptom Comparison**

- Grouped bar chart showed fever was more prevalent in male patients, while headache and fatigue were slightly higher in female patients.

These visualizations provided intuitive insights into the dataset and formed the backbone of the interactive dashboard developed for this project.

6: RESULTS & DISCUSSION

6.1 Analysis of Results

- **Symptom Patterns**

- Most frequent symptoms: Fatigue (62%), Fever (48%), Headache (41%), Cough (36%)

- Symptom combinations varied by age group, with younger individuals reporting fewer symptoms overall

- Patients with ≥3 symptoms had a higher likelihood of delayed recovery

- **Recovery Duration**

- Average recovery time: 18.6 days

- Fast recovery (<14 days): 42% of patients

- Delayed recovery (>28 days): 19% of patients

- Chronic conditions increased recovery time by an average of 9.2 days

- **Chronic Condition Influence**

- 32% of patients had at least one chronic illness (e.g., diabetes, asthma)

- These individuals showed longer recovery durations and higher symptom severity

- Strong correlation observed between chronic burden and recovery band classification

- **Gender-Based Insights**

- Fever was more prevalent in male patients

- Fatigue and headache were slightly more common among female patients

- Recovery duration showed no significant gender-based difference

- **Age Group Trends**

- Patients aged 50+ had the highest proportion of delayed recovery

- Younger age groups (≤30) recovered faster and reported fewer chronic conditions

6.2 Performance of Analysis Techniques

- Bar plots effectively highlighted symptom frequency and demographic distributions

- Heatmaps revealed strong correlations between chronic conditions and recovery time

- Pie charts provided intuitive breakdowns of chronic condition prevalence

- Grouped bar charts enabled clear gender-based comparisons

- Dashboard slicers enhanced interactivity, allowing users to filter by age, gender, and condition status

These results validate the effectiveness of Python-based visualization in uncovering meaningful health patterns and support the use of data dashboards for public health awareness and planning.

6.3 Results Comparison

- Individuals with chronic conditions took approximately 50% longer to recover than those without

- Patients aged 50+ had a 30% higher incidence of delayed recovery compared to younger age groups

- Symptom frequency in this dataset aligns with global post-COVID studies, but fatigue and headache were reported at slightly higher rates

- Compared to hospital-based studies, this dataset (community-level) showed milder severity levels but longer recovery durations

- Gender-based symptom distribution was consistent with existing literature, though fever prevalence in males was marginally higher than expected

7. CONCLUSION & FUTURE WORK

7.1 Summary of Findings

This project revealed key insights into post-COVID recovery patterns using Python-based data analysis and visualization:

- Fatigue, fever, and headache were the most frequently reported symptoms

- Individuals with chronic conditions experienced significantly longer recovery durations

- Age and symptom count were strong indicators of recovery speed

- Gender-based symptom differences were observed, though recovery time remained consistent across genders

- The interactive dashboard enabled intuitive exploration of symptom trends and demographic impacts

These findings highlight the importance of data-driven approaches in understanding post-COVID health outcomes and support the use of visual analytics for public health awareness.

7.2 Limitations of the Study

Several limitations should be acknowledged:

- The dataset was limited to a single source and may not represent broader geographic or clinical diversity

- Self-reported symptoms may introduce bias or inconsistencies

- No predictive modeling was performed; the study focused solely on descriptive analysis

- Severity levels were subjective and not clinically validated

- The absence of time-series data restricted longitudinal trend analysis

7.3 Future Scope & Enhancements

Future research directions include:

- Expanding the dataset to include hospital-based and regional health records

- Incorporating time-series data to study symptom progression and relapse

- Developing predictive models to classify recovery risk based on symptoms and conditions

- Enhancing the dashboard with real-time data integration and mobile accessibility

- Collaborating with healthcare providers to validate findings and improve data quality

- Applying machine learning techniques for symptom clustering and recovery forecasting

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